

KINEMATICS

MOTION IN ONE DIMENSION

IMPORTANT TERMS

1. MECHANICS

It is the branch of Physics, which deals with the study of motion of physical bodies.

Mechanics can be broadly classified into following branches

1.1 Statics

It is the branch of mechanics, which deals with the study of physical bodies at rest.

1.2 Kinematics

It is the branch of mechanics, which deals with study of motion of physical bodies without taking into account the factors, which causes motion.

1.3 Dynamics

It is the branch of mechanics, which deals with the study of motion of physical bodies taking into account the factors which causes motion.

2. REST AND MOTION

2.1 Rest

- An object is said to be at rest if it does not changes its position with respect to the surrounding.
- The white board in the classroom is at rest with respect to the classroom.

2.2 Motion

- An object is said to be in motion if it changes its position with respect to the surrounding.
- When we walk, run or ride a bike we are in motion with respect to the ground.

2.3 Rest and Motion are relative

Rest and motion depends upon the **observer**. The object in one situation may be at rest whereas the same object in another situation may be in motion.

The driver of a moving car is in motion with respect to an observer standing on the ground whereas, the same driver is at rest with respect to the man(observer) in the passengers seat.

3. WHILE STUDYING THIS CHAPTER

- We will treat the objects as **Point mass** object
- An object can be considered as a point mass object if during the course of motion it covers distances much greater than its own size.
- We shall confine ourselves to the study of rectilinear motion
 - Rectilinear motion is the study of motion of objects along a straight line.

4. POSITION, DISTANCE, DISPLACEMENT

4.1 Position

- Position of an object is always defined with respect to some reference point which we generally refer to as origin.
- To define the change in position we have two physical quantities.

4.2 Distance

- It is the actual path traversed by the body during the course of motion
- SI unit is 'm'
- Dimensions $[M^0L^1T^0]$

4.3 Displacement

- It is the difference between the final and initial positions of the object during the course of motion
- SI unit is 'm'
- Dimensions $[M^0L^1T^0]$